

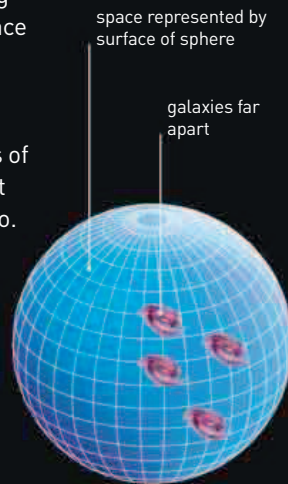
EXPANDING UNIVERSE

The first compelling piece of evidence in favour of the Big Bang theory was the discovery of the cosmic background radiation. This radiation was produced around 300,000 years after the Big Bang, and shows that the Universe was much smaller and hotter than it is now. The Universe's expansion has stretched the radiation, so that it is now mostly long-wavelength microwave radiation. Expanding 3-D space is best visualized as the growing surface of an inflating sphere. The relatively recent increase in the rate of expansion is caused by a poorly understood form of mass-energy called dark energy (see right and 1998). Measurements of the expansion and other key parameters suggest that the Big Bang occurred 13.8 billion years ago.



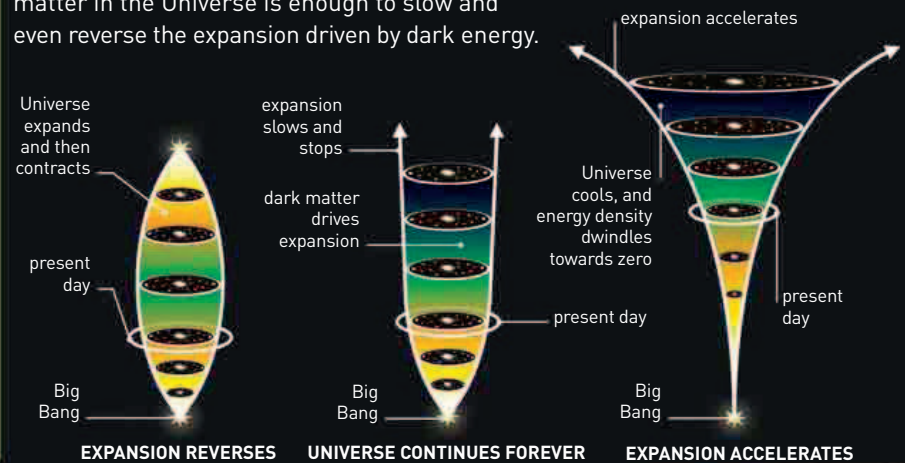
HUBBLE LAW

Edwin Hubble found that the further away a galaxy is, the faster it is receding. The mathematical relationship between a galaxy's distance and its speed is called Hubble's Law. It is best explained by the fact that space itself is expanding.



IN THE END

The Universe appears to be dominated by dark matter and dark energy—forms of matter and energy that have not been directly observed but whose existence is inferred from their gravitational influence and effect on the Universe's rate of expansion. If this is indeed the case, the fate of the Universe depends on whether the mutual gravitational attraction of observable and dark matter in the Universe is enough to slow and even reverse the expansion driven by dark energy.



FATE OF THE UNIVERSE

Three possible scenarios for the fate of the Universe are shown here. The gravitational influence of matter and dark matter could slow the expansion of the Universe, until it reaches a maximum. It is more likely that expansion will reverse or continue forever.

60 billion miles (100 billion km)

10^{13} k (18 trillion °F/ 10 trillion °C)

Protons and neutrons

The Universe continued to expand and cool, though more slowly, and fundamental particles called quarks combined into compound particles called baryons. The most important are protons and neutrons—these later formed the nuclei of atoms.

1 microsecond

10^{-6} seconds (1 millionth of a second)

1,000 light-years

10^6 k (180 million °F/100 million °C)

Opaque era

For the next 300,000 years, the temperature was too high for atoms to form. Charged particles such as protons and electrons constantly produced and absorbed photons (particles of electromagnetic radiation), making the Universe opaque.

200 seconds

100 million light-years

3,000k (4,900 °F/2,700 °C)

Matter era

As the Universe cooled, electrons began settling down into orbits around the atomic nuclei. The cosmic background radiation (see above) dates from this time. The Universe has continued to expand ever since.

300,000 years

helium-4 nucleus has two protons and two neutrons

photons produced by charged particles such as protons and electrons

Universe is opaque, as the movement of photons is constantly restricted

electrons become bound to nuclei, forming atoms

atoms will eventually clump together to form stars